

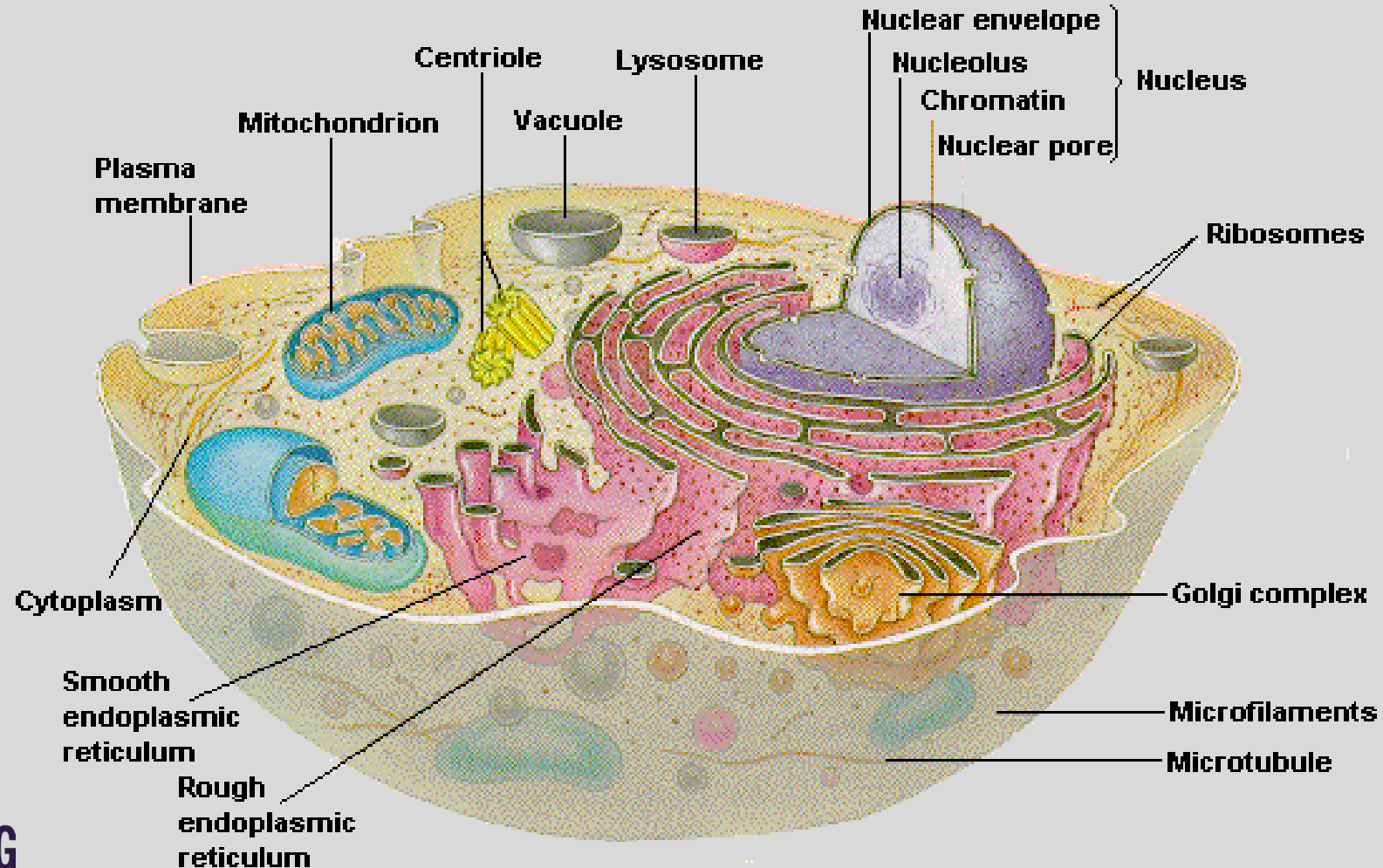


A practical example, protein localization

- Proteins are synthesized in the cytosol.
- Transported into different subcellular locations where they carry out their functions.
- Aim: To predict in what location a certain protein will end up!!!



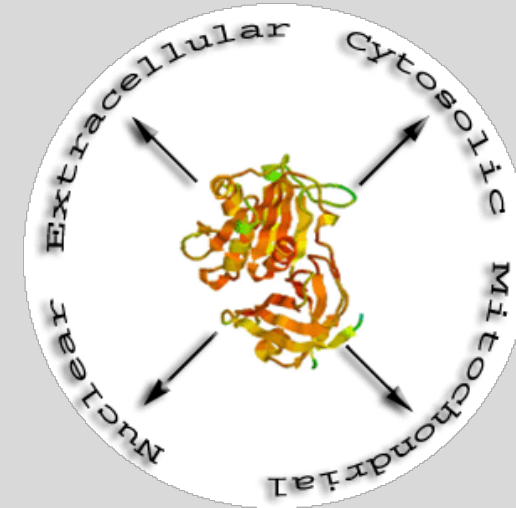
Subcellular Locations





Method

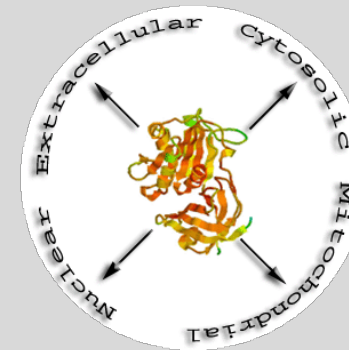
- Hypothesis: The amino acid composition of proteins from different compartments should differ.
- Extract proteins with know subcellular location from SWISSPROT.
- Calculate the amino acid composition of the proteins.
- Try to differentiate between: cytosol, extracellular, mitochondria and nuclear by using SVM





Input encoding

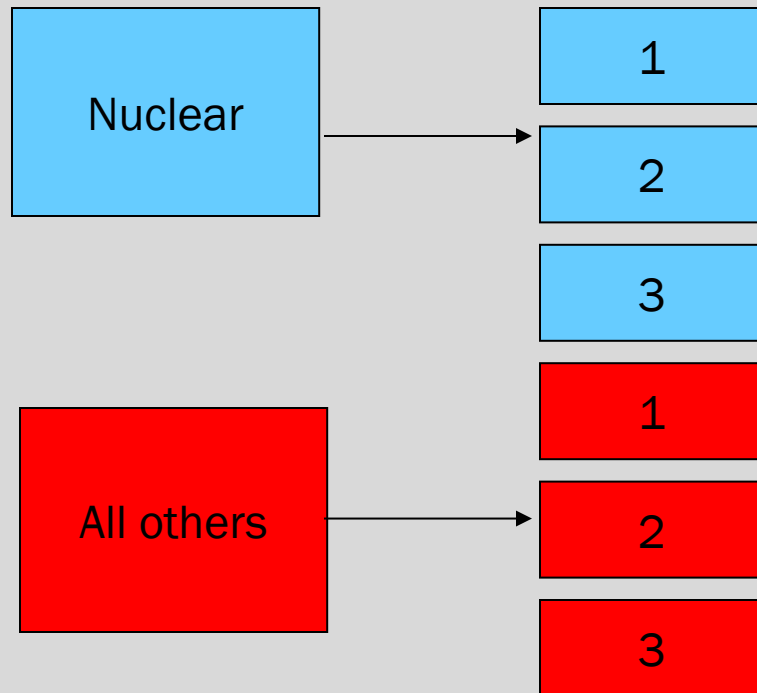
- Prediction of nuclear proteins:
- Label the known nuclear proteins as +1 and all others as -1.
- The input vector x_i represents the amino acid composition.
- Eg $x_i = (4.2, 6.7, 12, \dots, 0.5)$
(A, C, D, ..., Y)



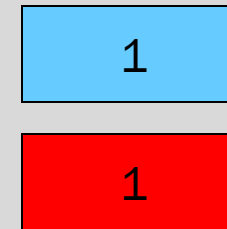


Cross-validation

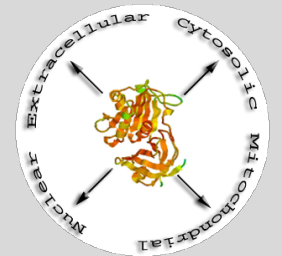
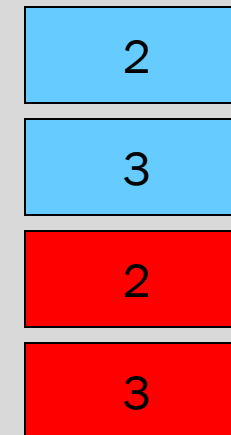
Cross validation: Split the data into n sets, train on $n-1$ set, test on the set left out of training.

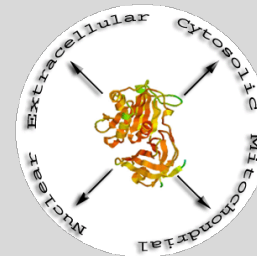


Test set

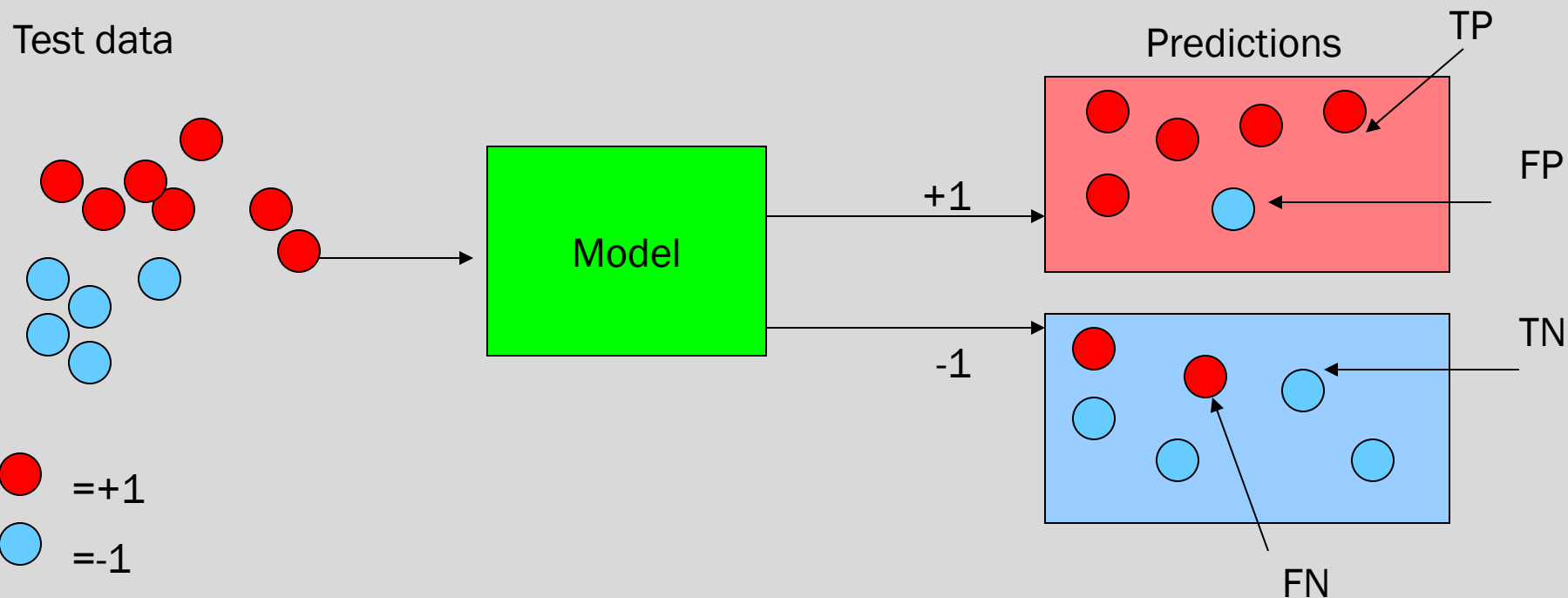


Training set





Performance measurements



SP = $TP / (TP + FP)$, the fraction of predicted +1 that actually are +1.

SE = $TP / (TP + FN)$, the fraction of the +1 that actually are predicted as +1.

In this case: $SP = 5 / (5 + 1) = 0.83$

$SE = 5 / (5 + 2) = 0.71$



A Cautionary Example

Image classification of tanks. Autofire when an enemy tank is spotted.

Input data: Photos of own and enemy tanks.

Worked really good with the training set used. In reality it failed completely.

Reason: All enemy tank photos taken in the morning. All own tanks in dawn.

The classifier could recognize dusk from dawn!!!!





References

<http://www.kernel-machines.org/>

<http://www.support-vector.net/>

AN INTRODUCTION TO SUPPORT VECTOR MACHINES

(and other kernel-based learning methods)

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